DERIVATIVE RULES QUIZ

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Name: Solutions

1. Compute the line tangent to

$$f(x) = 2x^3 - 3x^2 - 4x$$

at $x = 1$.

The line tangent to $f(x)$ at $x=1$ is

$$y - f(1) = f'(1)(x-1)$$

$$f'(x) = 6x^2 - 6x - 4 \Rightarrow f'(1) = 6(1)^2 - 6(1) - 4 = -4$$

$$f(1) = 2(1)^3 - 3(1)^2 - 4(1) = 2 - 3 - 4 = 2 - 7 = -5$$

$$\boxed{y + 5 = -4(x-1)}$$